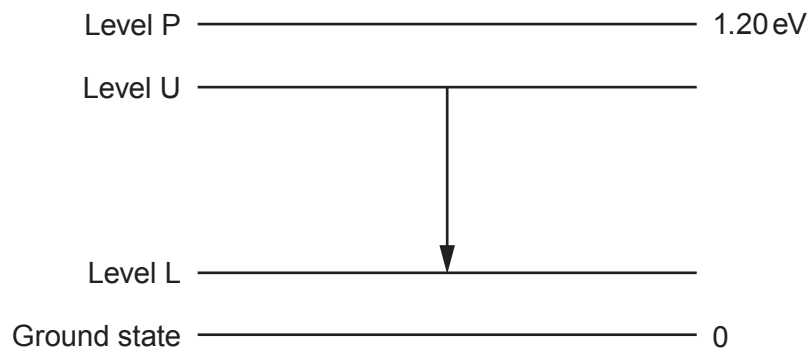


7. A simplified energy level diagram for the amplifying medium of a 4-level laser is shown below. The lasing transition occurs between level U and level L.



- (a) (i) The laser is pumped to create a population inversion between level U and level L. State what is meant by a population inversion for **this** laser. [1]

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- (ii) **Draw three arrows** on the diagram to show the transitions required for this population inversion to be sustained. [1]

- (iii) Explain why a population inversion is needed for light amplification to take place. [3]

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- (b) Determine whether or not visible light will be produced by the lasing transition from level U to level L, giving your reasoning. [3]

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